

When Is Model Simplification Safe? Consequence and Detectability for Common-Shape Weibull Series Systems under Masked Data

Alexander Towell

Department of Computer Science
Southern Illinois University Edwardsville
lex@metafunctor.com

Abstract

A reliability engineer modeling a Weibull series system must choose between a full model with a separate shape parameter for each component and a parsimonious reduced model that assumes a common shape. We show that the two questions a practitioner actually cares about, how much accuracy the simplification costs and whether the data can detect the simplification is wrong, are structurally decoupled. The reduced model’s prediction bias is a sample-size-independent population quantity that grows as $c_B CV^2$ in the shape coefficient of variation (with $c_B \approx 0.28$ for our system), while the likelihood ratio test’s detection power is governed by a noncentrality parameter that scales as $c_D n CV^2$ (with $c_D \approx 0.5$), so the detectable shape-CV threshold falls as $\approx 3.5/\sqrt{n}$. We make this precise with a delta-method bias expansion and a Le Cam local-power derivation, and we explain it with a structural result: at any common-shape point the common-shape constraint yields *exactly zero* first-order efficiency gain for system-level prediction. The same identity forces the misspecification bias to be second-order in heterogeneity (no linear term) and explains a counterintuitive finding, that at moderate sample sizes the full model has lower mean time-to-failure mean squared error than the reduced model *even when the common shape is exactly correct*. The practical consequence is a two-regime safety rule: for $n \gtrsim 1000$ the test rejects the reduced model before its bias becomes engineering-significant, and for $n \lesssim 500$ sampling error dominates the small specification bias, so the choice barely matters. All results hold under right-censored and masked failure data satisfying the standard conditions of the masked-data likelihood framework.

1 Introduction

Consider a turbine engine whose operation depends on bearings, seals, and blades arranged in series, so that any component failure grounds the engine. Field maintenance logs record the failure time and a short list of suspect components (the candidate set), but exact root-cause analysis requires costly teardown. Warranty programs further right-censor units still in service. The engineer faces a fundamental modeling choice: a full model with separate Weibull shape and scale parameters for each of m components ($2m$ parameters), or a reduced model that assumes a common shape parameter across all components ($m + 1$ parameters).

The full model is more flexible but harder to estimate from limited masked data. The reduced model is more parsimonious and, uniquely among single-shape reductions, renders the system life-time itself Weibull [1], enabling closed-form reliability metrics. But is the simplification safe? The bulk of the masked-data literature develops *estimators* for component reliability, by maximum likelihood [2, 3, 4, 5], by Bayesian methods [6], and for distributions beyond the Weibull [7, 8, 9, 10];

standard reliability texts [11, 12] treat the underlying lifetime models. The model-selection literature asks whether a test can *detect* a difference between candidate models [13, 14]. Neither answers the question a practitioner actually faces: when the reduced model is wrong, does it produce predictions wrong enough to change an engineering decision, and if so, will the data warn us in time?

1.1 The decoupling

We show that prediction consequence and statistical detectability are governed by different mechanisms that scale differently in the sample size n . Writing CV for the coefficient of variation of the component shapes, our central findings are:

- **Consequence is sample-size independent and quadratic.** The reduced model’s relative bias in system mean time-to-failure (MTTF) is a population quantity, $B \approx c_B CV^2$ with $c_B \approx 0.28$, that does not shrink with n . It stays below 1% through $CV \approx 18\%$.
- **Detectability accumulates with n .** The likelihood ratio test (LRT) has noncentrality $\approx c_D n CV^2$ with $c_D \approx 0.5$, so the smallest detectable CV falls as $\approx 3.5/\sqrt{n}$.

These are not two growth rates of the same quantity; they are two quantities with a shared geometric origin (Section 5). The practical upshot is a two-regime safety rule rather than a single “safe zone”: at large n the test reliably rejects the reduced model before the bias matters, and at small n the bias is dwarfed by sampling error, so the choice is second order either way.

1.2 Contributions

1. **A decoupling result with calibrated constants.** We derive the $O(CV^2)$ bias scaling by the delta method [15] and the $c_D n CV^2$ detection scaling by Le Cam’s local-alternative theory [16], and we calibrate both constants against simulation. This converts a single simulation study into formulas a practitioner can evaluate at any design point.
2. **A zero-first-order-gain proposition.** At any common-shape parameter point, the common-shape constraint provides *exactly zero* first-order asymptotic variance reduction for any functional of the system lifetime distribution (MTTF, reliability, quantiles). One identity (Section 5) yields two consequences: the misspecification bias has no linear term (hence $O(CV^2)$), and the finite-sample MSE ordering is decided at second order, where it favors the *full* model. This refutes the textbook intuition that parsimony reduces variance, for system-level prediction.
3. **An honest evaluation of pretest model selection.** The natural “fit both, test, pick one” procedure is a pretest estimator [17, 18]. We evaluate it (and its AIC and BIC analogues) and show it exhibits the classical pretest risk hump at $n = 100$ but is nearly costless for $n \geq 500$, exactly as the decoupling predicts.

1.3 Relation to focused model selection and pretesting

The closest framework is the focused information criterion (FIC) of Claeskens and Hjort [19], Hjort and Claeskens [20], which selects a model to minimize the estimated asymptotic MSE of a chosen “focus” functional, exactly the role MTTF plays here. FIC operates under *local* misspecification, where the parameter departs from the submodel at rate $O(1/\sqrt{n})$, so bias and variance enter at the same order. Our consequence analysis instead treats *fixed* misspecification, whose bias is $O(CV^2)$ and independent of n (the asymptotics of estimation under a misspecified parametric family go back

to Pascual [21] for the Weibull and White [22] in general); this is what makes the bias an engineering quantity rather than an asymptotic one. The two views meet in our zero-first-order-gain proposition, which implies that an FIC comparison would favor the full model essentially everywhere, recovering our recommendation by a different route. FIC and its model-averaging relatives [23] have been carried to Cox regression [24] but not, to our knowledge, to Weibull series systems or masked data; for the broader information-criterion view of model selection see Burnham and Anderson [25]. The pretest procedure of Section 6 is the classical pretest estimator [17, 18, 26, 27], and its small-sample risk hump and the non-uniformity of post-selection inference [28] are well understood in general; our contribution is to quantify them in the masked Weibull series setting. Tests of common Weibull shape across independent samples date to McCool [14, 29]; Hart [30] studies estimation under a common-shape Weibull competing-risks model assuming it holds, complementary to our analysis of what happens when it does not.

The paper is organized as follows. Section 2 sets up the model and data. Section 3 quantifies the consequence of misspecification. Section 4 characterizes detectability. Section 5 derives the decoupling and the zero-gain proposition. Section 6 evaluates pretest selection. Section 7 concludes.

2 Model Framework

2.1 Series System with Weibull Components

Consider a series system of m independent components, where the system fails when any component fails. Each component lifetime T_j follows a two-parameter Weibull distribution with shape $k_j > 0$ and scale $\lambda_j > 0$:

$$f_j(t; k_j, \lambda_j) = \frac{k_j}{\lambda_j} \left(\frac{t}{\lambda_j} \right)^{k_j-1} \exp \left\{ - \left(\frac{t}{\lambda_j} \right)^{k_j} \right\}, \quad t > 0. \quad (1)$$

The system lifetime $T = \min\{T_1, \dots, T_m\}$ has reliability function

$$R(t; \boldsymbol{\theta}) = \exp \left\{ - \sum_{j=1}^m \left(\frac{t}{\lambda_j} \right)^{k_j} \right\}, \quad (2)$$

with hazard $h(t; \boldsymbol{\theta}) = \sum_j (k_j/\lambda_j)(t/\lambda_j)^{k_j-1}$ and density $f = h \cdot R$, where $\boldsymbol{\theta} = (k_1, \lambda_1, \dots, k_m, \lambda_m)$ is the full parameter vector and $\text{MTTF}_j = \lambda_j \Gamma(1 + 1/k_j)$. The shape governs the failure mode: $k_j > 1$ gives an increasing hazard (wear-out), the regime of interest for aging components.

2.2 Masked and Censored Data

We observe n independent systems under two forms of incomplete information. *Right-censoring*: system i has indicator $\delta_i = 1$ if its failure time t_i is observed and $\delta_i = 0$ if it is still operating at the end of observation. *Masking*: for a failed system, instead of the true failed component K_i we observe only a candidate set $C_i \subseteq \{1, \dots, m\}$ of possible causes. We assume the three standard conditions of the masked-data likelihood framework [31]:

- (C1) **Containment**: the true cause is always in the candidate set, $\Pr(K_i \in C_i) = 1$.
- (C2) **Symmetry**: conditional on the failure time and on $K_i \in C_i$, the candidate set probability does not depend on which member of C_i is the true cause.
- (C3) **Independence**: the masking mechanism does not depend on $\boldsymbol{\theta}$.

Under C1–C3 the masking probabilities factor out of the likelihood and may be dropped, leaving

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \left[(1 - \delta_i) \log R(t_i; \boldsymbol{\theta}) + \delta_i \left(\log R(t_i; \boldsymbol{\theta}) + \log \sum_{j \in C_i} h_j(t_i; \boldsymbol{\theta}) \right) \right], \quad (3)$$

where we used $\sum_{j \in C_i} (h_j/h) f = R \sum_{j \in C_i} h_j$. Conditions C1–C3 are the standard assumptions of the masked competing-risks literature [32, 33, 34], under which the masked-data likelihood reduces to the candidate-set form above. We maximize (3) numerically with the L-BFGS-B algorithm [35] using analytic gradients; see Towell [31] for the framework and identifiability conditions.

2.3 The Common-Shape Model

The *common-shape model* (reduced model) constrains all components to share a single shape k , with parameter vector $\boldsymbol{\theta}_R = (k, \lambda_1, \dots, \lambda_m)$, reducing the parameter count from $2m$ to $m + 1$.

Property 2.1 (Weibull Closure [1]). *Let $T_1, \dots, T_m$ be independent with $T_j \sim \text{Weibull}(k, \lambda_j)$. Then $T = \min\{T_1, \dots, T_m\} \sim \text{Weibull}(k, \lambda_s)$ with $\lambda_s = (\sum_{j=1}^m \lambda_j^{-k})^{-1/k}$.*

Remark 2.1 (Uniqueness). *Equality of all shapes is not merely sufficient for a Weibull system lifetime; it is necessary. The system cumulative hazard is $\sum_j (t/\lambda_j)^{k_j}$, a sum of power functions t^{k_j} . By the linear independence of distinct powers, this equals a single power $(t/\lambda_s)^{k_s}$ for all $t > 0$ only if $k_1 = \dots = k_m = k_s$. Thus the common-shape model is the unique reduction of the full model that preserves the Weibull property of the system, which is what makes it the natural parsimony boundary.*

The practical question is not whether closure holds exactly, but whether the common-shape model’s predictions are adequate when shapes are only *approximately* equal. In well-designed systems components age similarly ($k_j \approx k$) but differ in durability (λ_j), so the common-shape model is physically motivated.

2.4 Baseline System and Simulation Design

We measure departure from shape homogeneity by the coefficient of variation $\text{CV} = \text{sd}(k_1, \dots, k_m) / \text{mean}(k_1, \dots, k_m)$. Throughout, we use a 5-component baseline from Guo et al. [36] with $\text{CV} \approx 4\%$ (Table 1).

Table 1: Baseline 5-Component Series System

Component j	Shape k_j	Scale λ_j	MTTF $_j$
1	1.2576	994.37	≈ 913
2	1.1635	908.95	≈ 859
3	1.1308	840.11	≈ 799
4	1.1802	940.13	≈ 886
5	1.2034	923.16	≈ 866

Unless stated otherwise, simulations use masking probability $p = 0.215$ (each non-failed component enters the candidate set independently with probability p , satisfying C1–C3) and censoring at the $q = 0.825$ system-lifetime quantile, with 500 replications per condition. To vary shape heterogeneity we generate m shapes equally spaced about a mean of 1.18 at a target CV; because the shapes are discrete the realized CV differs slightly from the target (target CV = 10% yields actual CV $\approx 13.7\%$). All tables report the actual CV; prose uses approximate values for readability. All

fits use the reference `wei.series.md.c1.c2.c3` implementation [37] with analytic gradients; the optimizer settings and the handling of non-convergent replications are detailed in Appendix A. A separate analysis of the masking-invariance and Fisher-information structure of the homogeneous model appears in Towell [38].

3 Consequence Analysis: Does Misspecification Matter?

We fit the reduced model to data generated from heterogeneous-shape systems and measure the resulting prediction error. The key estimand is the system MTTF, $\text{MTTF} = \int_0^\infty R(t; \theta) dt$; we also report the reliability $R(t)$ at $t = \text{MTTF}/2$, MTTF , 2MTTF . The relative bias of an estimate $\hat{M}$ is $(\hat{M} - M_{\text{true}})/M_{\text{true}}$.

3.1 Specification bias is quadratic and sample-size independent

Table 2 reports the reduced model’s relative MTTF bias across shape CV and sample size, alongside the population (limiting, $n \rightarrow \infty$) specification bias obtained by projecting the heterogeneous truth onto the reduced family in Kullback–Leibler divergence (Appendix A). Three patterns stand out.

Table 2: Relative MTTF Bias (%) of the Common-Shape Model by Actual Shape CV. The population column is the $n \rightarrow \infty$ specification bias; the finite- n columns add finite-sample estimation bias, conspicuous at CV = 0 where specification bias is zero.

CV (%)	population	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$
0.0	+0.00	+0.36	−0.10	+0.04	−0.04
2.7	+0.02	+0.30	−0.09	−0.12	+0.04
5.5	+0.08	+0.80	+0.20	+0.07	+0.14
8.2	+0.18	+0.74	+0.45	+0.09	+0.19
11.0	+0.32	+1.10	+0.62	+0.24	+0.22
13.7	+0.52	+0.85	+0.89	+0.50	+0.55
20.5	+1.31	+1.65	+1.49	+1.29	+1.32
27.4	+2.94	+3.22	+2.93	+2.52	+2.59
41.1	—	+10.5	+10.1	+9.8	+9.3

First, the specification bias is small and positive (the common-shape fit overestimates the weaker components, inflating MTTF) and stays below 1% through CV $\approx 18\%$. Second, it is essentially independent of n : a 50-fold increase from $n = 100$ to $n = 5000$ leaves the bias at CV $\approx 27\%$ near +2.6%, confirming a specification effect rather than a finite-sample artifact. Third, the population curve makes the decomposition explicit: the +0.36% in the $n = 100$ column at CV = 0, where there is *no* specification bias, is pure finite-sample estimation bias, additive to and distinct from the specification bias that the rest of the column conflates with it. Figure 1 shows the population bias is quadratic in CV with no linear term, $B \approx 0.28 \text{ CV}^2$, a fact we derive in Section 5. The reliability predictions behave similarly: at $n = 1000$, $R(t)$ relative bias stays below 1% in magnitude through CV $\approx 14\%$ at all three horizons, with the deep tail $R(2\text{MTTF})$ the most sensitive (Table 3).

3.2 The full model can have lower MSE, even when shapes are equal

A counterintuitive finding emerges at CV = 0, where the reduced model is *correctly specified*. Table 4 shows that the full model has lower MTTF root-mean-squared error at moderate sample sizes, with the gap statistically significant by a paired test at $n = 100$. The mechanism is variance, not bias: at $n = 100$ the constrained estimator’s MTTF variance is about 20% larger than the unconstrained

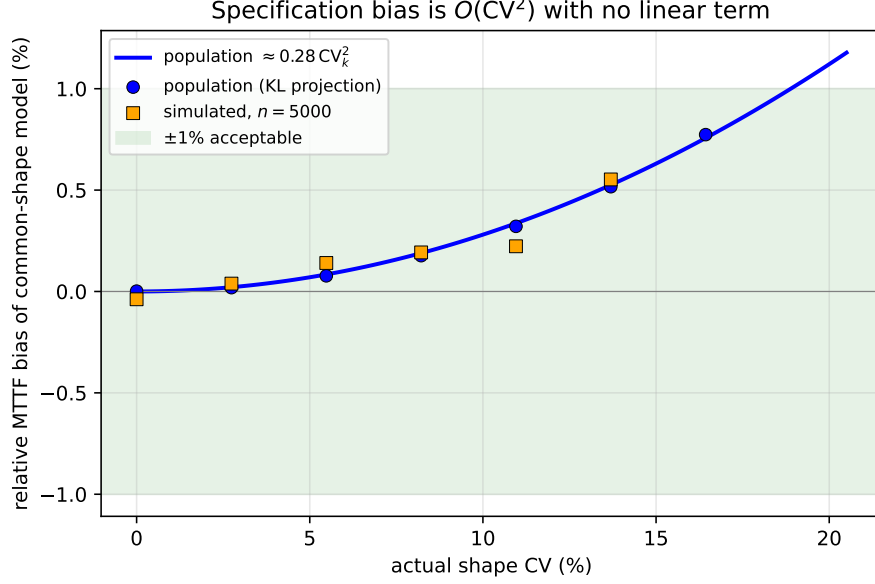


Figure 1: Population (limiting) specification bias of the common-shape model versus shape CV, from the Kullback–Leibler projection under masking and censoring (blue), with simulated $n = 5000$ bias (orange) and the quadratic fit $0.28 CV^2$. The bias vanishes to second order: there is no linear term (Section 5), so it stays inside the $\pm 1\%$ engineering band through $CV \approx 18\%$.

Table 3: Relative Bias (%) of Common-Shape $R(t)$ Predictions at $n = 1000$

CV (%)	$R(\text{MTTF}/2)$	$R(\text{MTTF})$	$R(2 \text{ MTTF})$
0.0	+0.03	−0.01	−0.08
5.5	−0.09	−0.15	+0.19
11.0	−0.27	−0.41	+0.83
13.7	−0.38	−0.52	+1.65
20.5	−0.86	−1.04	+3.81
27.4	−1.53	−1.82	+6.08

estimator's, while the squared bias of both is negligible (and what little there is favors the reduced model). The variance inflation shrinks with n and is gone by $n = 5000$. This is not a small-sample curiosity to be explained away; it is the finite- n shadow of an exact structural fact, that the common-shape constraint buys *zero* first-order variance reduction for system MTTF, which we prove in Section 5. The traditional argument that parsimony reduces variance holds for the shape parameters themselves but not for the nonlinear system functional built from them.

Table 4: MTTF Error at CV = 0 (reduced model correctly specified). RMSE as % of true MTTF; paired t on per-replication squared-error difference (negative favors the full model); variance ratio of the two MTTF estimators.

n	RMSE full (%)	RMSE reduced (%)	paired t	Var _R /Var _F
100	8.6	9.2	−3.65	1.20
500	4.3	4.3	−1.80	1.03
1000	3.0	3.1	−1.84	1.02
5000	1.3	1.3	+0.24	1.00

4 Detectability: Likelihood Ratio Testing

We test $H_0 : k_1 = \dots = k_m$ with the likelihood ratio statistic

$$\Lambda = -2(\ell(\hat{\theta}_R) - \ell(\hat{\theta}_F)), \quad (4)$$

asymptotically χ^2_{m-1} under H_0 [39]; the null is interior (the common shape is positive), so the standard regularity conditions hold and there is no boundary correction.

4.1 Type I error and power

Under perfect homogeneity (CV = 0) the rejection rate ranges from 4.6% to 6.8% across n from 100 to 10,000. With 500 replications the binomial standard error is about 1%, so every value is consistent with the nominal $\alpha = 0.05$; the test is well calibrated. Table 5 reports power as a function of shape CV.

Table 5: LRT Rejection Rate by Shape CV and Sample Size (500 replications; binomial SE ≤ 0.022)

CV (%)	$n = 100$	$n = 500$	$n = 1000$	$n = 5000$	$n = 10000$
0.0	0.054	0.056	0.046	0.068	0.046
2.7	0.048	0.062	0.052	0.186	0.268
5.5	0.075	0.072	0.130	0.541	0.887
8.2	0.074	0.126	0.266	0.936	0.996
11.0	0.088	0.252	0.494	0.996	1.000
13.7	0.129	0.385	0.690	1.000	1.000
20.5	0.200	0.790	0.968	1.000	1.000

At CV $\leq 5\%$ the test has little power at practical sample sizes ($n \leq 1000$); even at $n = 10,000$ the rejection rate at CV = 2.7% is only 0.27. Combined with the consequence analysis, where bias is negligible in this regime, the case for the common-shape model is strong precisely where the test cannot reject it. Appendix B examines how masking, censoring, and system size affect power: masking has the largest effect ($6\times$ power reduction from $p = 0.05$ to 0.70), then censoring ($2.5\times$), then component count, with no inflation of Type I error. Appendix C compares the LRT with AIC and BIC.

5 Theoretical Analysis: Why Bias and Detection Decouple

The empirical patterns of Sections 3 and 4, quadratic n -free bias, $\sqrt{n}$ -shrinking detectability, and a full model that wins at CV = 0, all follow from a single geometric fact about the common-shape constraint. We state it, prove its consequences, and calibrate the constants.

5.1 A conjugacy identity

Work at a common-shape point θ_0 with $k_1 = \dots = k_m = k$ and arbitrary scales, and reparameterize each component by (k_j, η_j) with $\eta_j = \lambda_j^{-k_j}$, so the system cumulative hazard is $H(t) = \sum_j \eta_j t^{k_j}$, which at the common-shape point is st^k with $s = \sum_j \eta_j$. Let g be any smooth functional of the system lifetime distribution (MTTF, $R(t)$ at a fixed t , a quantile), let I be the Fisher information at θ_0 , and let $V = \{v : \text{the } k\text{-block of } v \text{ is constant across components}\}$ be the reduced model's tangent space embedded in the full $2m$ -dimensional space.

Proposition 5.1 (Conjugacy identity). *At any common-shape point, $I^{-1}\nabla g \in V$; equivalently, ∇g is I -conjugate to the shape-contrast directions, $\nabla g^\top I^{-1} A^\top = 0$, where A is the $(m-1) \times 2m$ contrast operator whose rows encode $k_l - k_{l+1} = 0$.*

Proof sketch. Two structural facts close the argument in a two-dimensional subspace. (i) *Gradient structure.* Because g depends on θ only through the system law, and at the common-shape point the cumulative hazard is st^k , g depends on η only through the sum s ; hence $\partial g / \partial \eta_l = c_\eta$ is the same for all l , while $\partial H / \partial k_l = \eta_l t^k \log t$ gives $\partial g / \partial k_l = c_k \eta_l$. So $\nabla g = (c_\eta \mathbf{1}, c_k \eta)$ lies in $P := \text{span}\{(\mathbf{1}, \mathbf{0}), (\mathbf{0}, \eta)\}$. (ii) *Information structure.* A direct computation of the complete-data score outer products at the common-shape point gives

$$I_{\eta\eta} = \frac{1}{s} \text{diag}(\eta^{-1}), \quad I_{\eta k} = \frac{1}{k} (c_1 I_m + \kappa \mathbf{1} \eta^\top), \quad I_{kk} = \frac{1}{k^2} (e_2 \text{diag}(\eta) + \psi \eta \eta^\top)$$

for scalars c_1, κ, e_2, ψ depending only on s . These four blocks map $U := \text{span}\{(\eta, \mathbf{0}), (\mathbf{0}, \mathbf{1})\}$ into P . Since $\dim U = \dim P = 2$ and I is invertible, $I : U \rightarrow P$ is a bijection, so $I^{-1} : P \rightarrow U$, and as $\nabla g \in P$ we get $I^{-1}\nabla g \in U$, every member of which has a constant k -block. The full computation, including the masked, censored case, is in Appendix D. $\square$

Under masking and censoring the gradient structure is unchanged, and the four block proportionalities the proof needs continue to hold; we verify this to Monte Carlo precision for the baseline design (3×10^6 simulated observations), and the resulting first-order variance gain is below 10^{-5} in every configuration tested (Appendix D).

5.2 Consequence 1: zero efficiency gain and the MSE ordering

By the theory of constrained maximum likelihood [40, 41], imposing a true constraint changes the asymptotic variance of g from $\text{var}_F = \nabla g^\top I^{-1} \nabla g$ to $\text{var}_R = \text{var}_F - \nabla g^\top I^{-1} A^\top (A I^{-1} A^\top)^{-1} A I^{-1} \nabla g$. The subtracted gain vanishes exactly when $A I^{-1} \nabla g = 0$, which is the conjugacy identity. Hence:

Corollary 5.1. *Imposing the (true) common-shape constraint yields exactly zero first-order asymptotic variance reduction for any system-level functional, at any scale configuration.*

So parsimony buys no first-order efficiency for system-level prediction, and the finite- n MSE ordering is decided at second order. The simulations of Section 3.2 show the second-order effect favors the full model: the pooled common-shape error is coherent across components (one shape error shifts all m cumulative hazards the same way), whereas the full model's m shape errors partially diversify inside the nonlinear system functional. This is why $\text{Var}_R > \text{Var}_F$ at $n = 100$ and why the gap vanishes as n grows, exactly as Corollary 5.1 requires.

5.3 Consequence 2: the bias has no linear term

Let the true shapes be $k_l = \bar{k} + \delta_l$ with $\sum_l \delta_l = 0$, so $\text{CV} \propto \|\delta\|$. The reduced model's pseudo-true parameter is the KL projection of the truth onto V , and to first order it equals the I -orthogonal projection of the parameter shift onto V . The MTTF bias $B(\delta) = g(\theta_R^*(\delta)) - g(\theta_{\text{true}}(\delta))$ then has derivative $\partial B / \partial \delta_j = \nabla g \cdot (P_V^I e_{k_j} - e_{k_j})$ at $\delta = 0$. The vector $r = P_V^I e_{k_j} - e_{k_j}$ is I -orthogonal to V , and since $I^{-1} \nabla g \in V$ by Proposition 5.1, $\nabla g \cdot r = (I^{-1} \nabla g)^\top I r = 0$.

Corollary 5.2. *The misspecification bias of any system-level functional has no linear term in the shape heterogeneity: $B(\delta) = O(\|\delta\|^2) = O(\text{CV}^2)$.*

The same identity that nullifies the variance gain nullifies the linear bias. The deterministic KL projection confirms it: the population bias at $\text{CV} = 0$ is +0.002% (numerically zero), forward and reversed shape perturbations give identical bias (the bias is even in δ), and the leading coefficient is $c_B \approx 0.28$ (Figure 1).

5.4 Detectability and the decoupling

The detection side is governed by the same matrix $AI^{-1}A^\top$ but enters differently. Under local alternatives $k_l = \bar{k} + \delta_l$ with $\|\delta\| = O(n^{-1/2})$, the LRT statistic is asymptotically noncentral χ_{m-1}^2 with noncentrality $\text{nep} = (A\delta)^\top (AI^{-1}A^\top)^{-1} (A\delta)$ [16], which scales as $c_D n \text{CV}^2$. Fitting against the simulations gives $c_D \approx 0.5$ (the per-observation theoretical value is 0.49), stable across $n \in [10^3, 10^4]$ and $\text{CV} \in [2.7\%, 27\%]$. Figure 2(a) shows that plotting power against $n \text{CV}^2$ collapses every (n, CV) condition onto a single noncentral- χ^2 curve, and Figure 2(b) shows the resulting detectable threshold falling as $\text{CV}_{50} \approx 3.5/\sqrt{n}$ (fitted log-log slope -0.47 versus the theoretical -0.5).

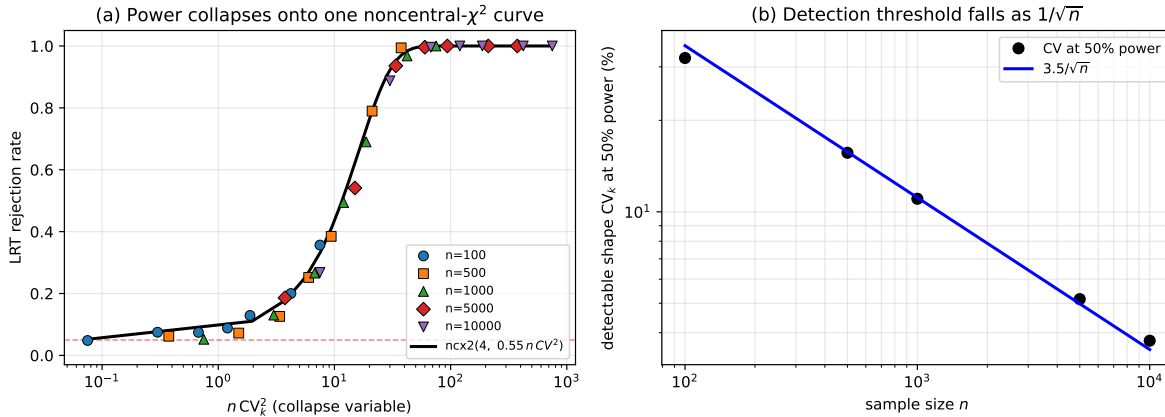


Figure 2: (a) The LRT power for every sample size and heterogeneity level collapses onto one noncentral- χ^2 curve when plotted against $n \text{CV}^2$, confirming noncentrality $\approx 0.5 n \text{CV}^2$. (b) The detectable shape-CV threshold (50% power) falls as $\approx 3.5/\sqrt{n}$.

The decoupling is now transparent. Bias is a population quantity, $O(\text{CV}^2)$ and free of n (Corollary 5.2); detectability accumulates with information, $O(n \text{CV}^2)$. They share the shape geometry but scale differently in n . Two practical regimes follow:

- $n \gtrsim 1000$ (**detection-led**): the detectable threshold $\text{CV}_{50} \approx 3.5/\sqrt{n}$ sits well below the $\text{CV} \approx 18\%$ bias boundary, so the test rejects the reduced model before its bias becomes engineering-significant.

- $n \lesssim 500$ (**noise-dominated**): the test rarely rejects, but the specification bias ($\leq 3\%$ through $CV \approx 27\%$) is dwarfed by sampling error (RMSE 4 to 10%), so the model choice is second order regardless.

At very large n the test may reject even trivially small heterogeneity, but Corollary 5.1 shows switching to the full model then costs nothing to first order. This is the structural answer behind the FIC discussion of Section 1.3: an MSE-focused comparison favors the full model essentially everywhere, while the closed-form bias and power formulas above serve the engineering decision directly at any fixed n .

6 Model Selection in Practice

Faced with data from an unknown system, the natural procedure is a pretest estimator [17, 18]: fit both models, compute Λ , use the reduced model if $\Lambda < \chi_{m-1, 1-\alpha}^2$ and the full model otherwise. This is not a new method, and its risk properties are governed by the classical pretest theory; our question is how that theory plays out here, given the decoupling.

6.1 Strategies and results

We compare always-full, always-reduced, and pretest selection by the LRT, and for completeness the AIC and BIC pretest variants (reduced if Λ is below $2(m-1)$ and $(m-1)\log n$, respectively), on MTTF RMSE across CV from 0 to 27% at $n \in \{100, 500, 1000\}$. Table 6 reports representative results.

Table 6: MTTF RMSE (% of true MTTF) by Strategy. “Sel. R.” is the LRT pretest’s reduced-selection rate.

CV (%)	n	Full	Reduced	LRT	AIC	BIC	Sel. R. (%)
0.0	100	9.3	10.1	10.0	10.0	10.1	93
	500	4.6	4.7	4.7	4.7	4.7	95
	1000	3.0	3.0	3.0	3.0	3.0	94
13.7	100	9.5	10.4	10.3	10.2	10.4	89
	500	4.3	4.5	4.4	4.4	4.5	60
	1000	3.0	3.2	3.0	3.0	3.1	30
27.4	100	9.9	12.8	11.6	11.1	12.2	65
	500	4.6	5.6	4.7	4.7	5.2	1
	1000	3.3	4.4	3.3	3.3	3.4	0

The LRT pretest tracks the better static strategy. Its RMSE overhead relative to always-full is at most 2.4% at $n = 500$ and $n = 1000$, but reaches 18% at $n = 100$ and $CV \approx 27\%$: this is the classical pretest risk hump [18], occurring where the test lacks the power to switch away from the now-biased reduced model. The AIC pretest behaves almost identically (its more liberal threshold helps slightly at small n), while the BIC pretest is worst at small n (overhead up to 24%): because BIC almost never selects the full model, it behaves like always-reduced and inherits its bias at high CV.

6.2 Why the pretest is nearly costless at $n \geq 500$

The decoupling explains the pattern directly. Where the pretest’s “wrong” choices occur (moderate CV, where the test lacks power) the reduced model’s bias is still small by Corollary 5.2, so misselection is cheap; where the bias is large the test has ample power to switch. The risk hump survives

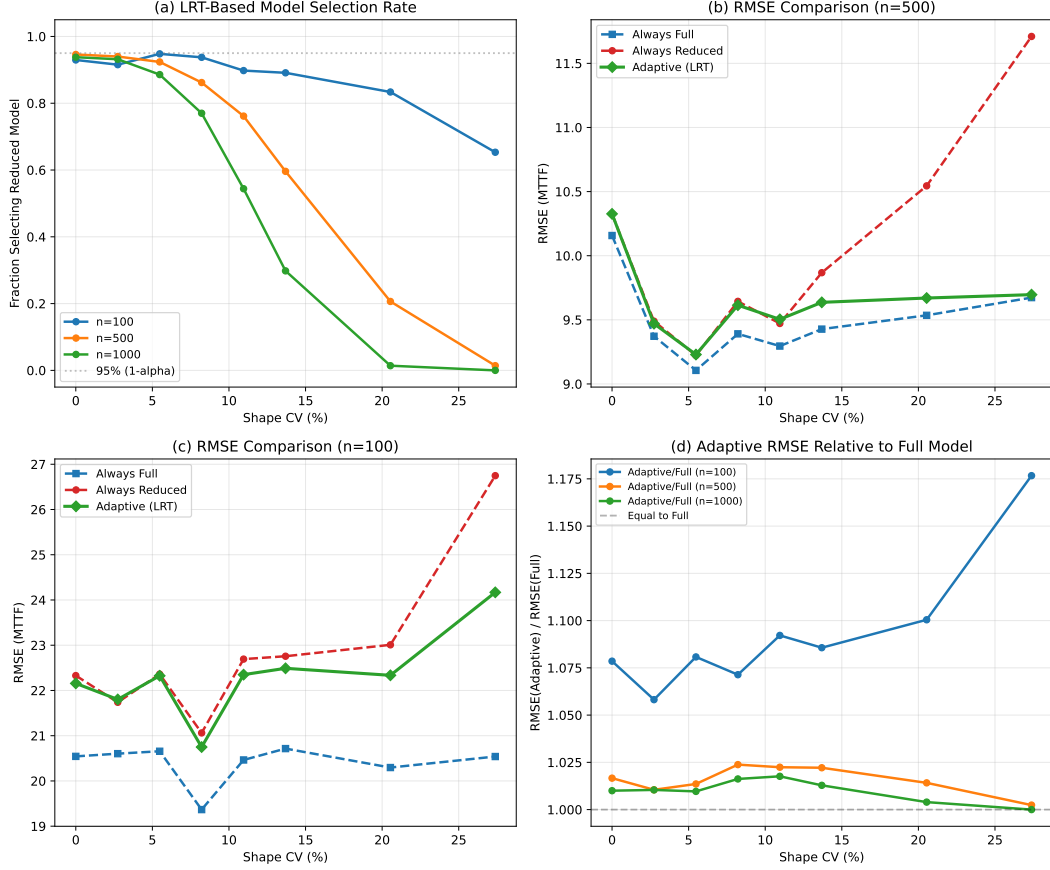


Figure 3: Pretest model selection. (a) Selection rate: the LRT pretest selects the reduced model near 95% at CV = 0 and transitions to the full model as heterogeneity grows, the transition shifting left with n . (b)–(c) RMSE at $n = 500$ and $n = 100$: the pretest tracks the better static strategy. (d) Relative RMSE overhead: under 2.5% at $n = 500$, 6 to 18% at $n = 100$.

only at $n = 100$, where neither regime of the two-regime rule has fully taken hold. Since both fits are already computed to form Λ , the pretest adds no estimation cost, and at $n \geq 500$ it is within 2.5% of always knowing the right model.

7 Conclusion

We asked when a reliability engineer can safely simplify a Weibull series model to a common shape, and answered it structurally. Misspecification consequence and statistical detectability are decoupled: the reduced model’s prediction bias is a sample-size-independent population quantity, $O(CV^2)$ with coefficient ≈ 0.28 , while the likelihood ratio test’s detectability scales as $\approx 0.5 n CV^2$, so the detectable threshold falls as $3.5/\sqrt{n}$. A single conjugacy identity drives both halves: at any common-shape point the constraint is invisible to first order for system-level prediction, which forces the bias to be second-order in heterogeneity and shows that the common-shape constraint provides zero first-order variance gain. The latter explains a finding that contradicts the parsimony-reduces-variance intuition: at moderate sample sizes the full model has lower MTTF MSE than the reduced model even when the common shape is exactly correct.

For practice this yields a two-regime safety rule. For $n \gtrsim 1000$ the LRT rejects the reduced model before its bias becomes engineering-significant; for $n \lesssim 500$ the specification bias is dominated by sampling error, so the choice is second order. The natural pretest procedure inherits this: it is within 2.5% of the better static strategy for $n \geq 500$, with a residual risk hump at $n = 100$ that is the classical pretest phenomenon, not a defect of the masked-data setting. The case for the common-shape model rests on Weibull closure and engineering-adequate predictions, never on statistical efficiency.

Limitations. Our calibrated constants ($c_B \approx 0.28$, $c_D \approx 0.5$, $CV_{50} \approx 3.5/\sqrt{n}$) are derived for the baseline 5-component system with equally spaced shapes; the scaling laws ($O(CV^2)$, $O(nCV^2)$) and the zero-gain proposition hold generally, but the coefficients should be recomputed for a given system from the formulas of Section 5. Real systems may exhibit clustered or asymmetric heterogeneity rather than the symmetric spread studied here. The masked-case proof of Proposition 5.1 is reduced to four block proportionalities verified numerically; a fully analytic masked-data proof is left open.

Future directions. Extending the decoupling to non-Weibull families (where closure may fail), to k -out-of- m systems, and to the focused-information-criterion machinery under fixed rather than local misspecification would broaden the reach of the structural result.

A Simulation Details

All fits use L-BFGS-B [35] with analytic gradients. Full-model fits use `parscale` equal to the parameter vector; reduced-model fits add a Nelder–Mead polishing step from the L-BFGS-B solution because the common-shape likelihood surface is mildly multimodal at high heterogeneity. Non-convergent full-model fits and the rare replications with $\Lambda < 0$ (an optimizer-failure indicator) are excluded: fewer than 3/500 per condition near $CV = 0$, rising to about 13/500 at $CV \approx 41\%$, so the headline results (which live at $CV \leq 27\%$) are essentially unaffected. The population specification bias of Table 2 and Figure 1 is computed without simulation, by minimizing the expected reduced-model negative log-likelihood (a deterministic integral under the true masking and censoring law) over the reduced parameters and evaluating the resulting MTTF; this gives the exact $n \rightarrow \infty$ pseudo-true bias and isolates specification from estimation error.

B Data Quality Effects on LRT Power

We varied masking probability p , censoring quantile q , and component count m for the baseline system (Table 7, $n = 5000$). Masking has the largest effect, censoring the next, and larger systems have lower power (failure information is spread across more parameters and the critical value grows). No factor inflates Type I error.

Table 7: Effect of Data Quality on LRT Power ($n = 5000$, Baseline)

Factor	Range	Power Range	Effect
Masking p	0.05–0.70	48% $\rightarrow$ 8%	$6\times$ reduction
Censoring q	0.50–1.00	17% $\rightarrow$ 43%	$2.5\times$ increase
Components m	2–8	77% $\rightarrow$ 16%	$5\times$ reduction

C Comparison with Information Criteria

Figure 4 compares the LRT with AIC [42] and BIC [43] as selectors. Under $CV = 0$ the LRT is well calibrated (4.6 to 6.8%); AIC is liberal (8.2 to 12.4%), and because its penalty does not grow with n its false-positive rate does not vanish but converges to $\Pr(\chi_{m-1}^2 > 2(m-1)) = 9.2\%$ for $m = 5$; BIC is over-conservative ($\leq 0.2\%$), unsuitable for detecting the subtle heterogeneity of a well-designed system. This is consistent with the pretest comparison of Section 6, where BIC’s reluctance to select the full model makes it behave like the always-reduced strategy.

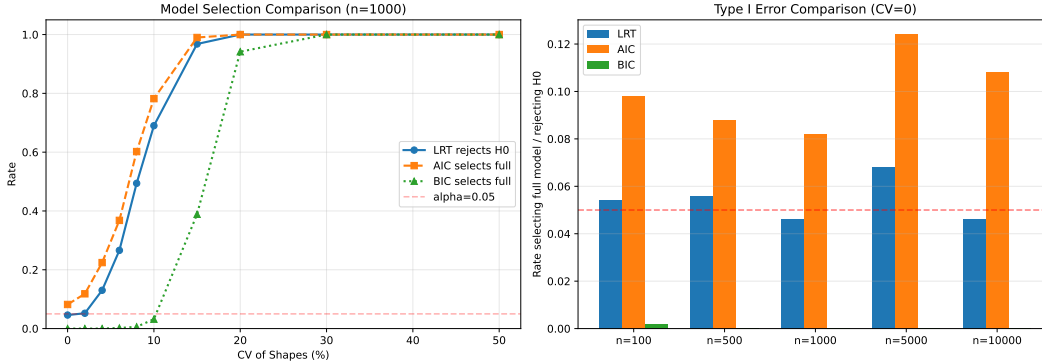


Figure 4: Model selection rates for LRT ($\alpha = 0.05$), AIC, and BIC versus shape CV.

D Proof of the Conjugacy Identity

Complete data. At the common-shape point, with $W = T^k \sim \text{Exponential}(s)$ and failed component J independent of W with $\Pr(J = l) = \eta_l/s$, the complete-data scores are $\partial\ell/\partial\eta_l = \mathbf{1}\{l = J\}/\eta_l - t^k$ and $\partial\ell/\partial k_l = k^{-1}[(1 + \log W)\mathbf{1}\{l = J\} - \eta_l W \log W]$. Taking expectations of the score outer products gives the blocks quoted in the proof sketch of Section 5, with $c_1 = (1 + \mathbb{E} \log W)/s$, $\kappa = \mathbb{E}[W^2 \log W] - (2\mathbb{E}[W \log W] + \mathbb{E}[W])/s$, $e_2 = \mathbb{E}[(1 + \log W)^2]$, and $\psi = \mathbb{E}[W^2(\log W)^2] - 2\mathbb{E}[(1 + \log W)W \log W]/s$. Writing $\nabla g = (c_\eta \mathbf{1}, c_k \eta)$ and seeking $v = (b\eta, a\mathbf{1}) \in U$ with $Iv = \nabla g$ reduces, using $\eta^\top \mathbf{1} = s$, to a 2×2 linear system in (a, b) that is solvable whenever the scalar $D = (e_2/s + \psi s) - s(c_1 + \kappa s)^2$ is nonzero (equivalently, I is nonsingular on the relevant subspace). The unique solution has constant k -block $a\mathbf{1}$, proving $I^{-1}\nabla g \in V$.

Masked and censored data. The gradient structure $\nabla g = (c_\eta \mathbf{1}, c_k \boldsymbol{\eta})$ is unchanged, since g remains a functional of the system law. The proof needs only that I map U into P , i.e. the four proportionalities $I_{\eta\eta} \boldsymbol{\eta} \propto \mathbf{1}$, $I_{k\eta} \boldsymbol{\eta} \propto \boldsymbol{\eta}$, $I_{\eta k} \mathbf{1} \propto \mathbf{1}$, $I_{kk} \mathbf{1} \propto \boldsymbol{\eta}$. We verified these for the baseline design ($p = 0.215$, $q = 0.825$, $m = 5$) by Monte Carlo, computing I from 3×10^6 simulated masked, censored observations: the relative deviations from proportionality are at the 5×10^{-4} Monte Carlo floor, and the end-to-end first-order variance gain for MTTF and $R(t)$ is below 10^{-5} . A fully analytic treatment requires propagating the candidate-set averaging through the block expectations and is left open.

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